

第十三章作业

13.7

对左侧网孔，

$$(2+j6)I_1 - jI_2 = 24\angle 0^\circ \quad (1)$$

对右侧网孔，

$$-jI_1 + (2+j3)I_2 = 0 \quad (2)$$

则有：

$$jI_1 = (2+j3)I_2$$

$$I_1 = \frac{2+j3}{j}I_2 = -j(2+j3)I_2$$

$$I_1 = (3-j2)I_2, \text{ 代入 (1) 得:}$$

$$(2+j6)(3-j2)I_2 - jI_1 = 24$$

$$\text{即 } (18+j13)I_2 = 24$$

$$\text{解得 } I_2 = \frac{24}{18+j13}$$

$$\text{则 } V_0 = 1 \times I_2 = \frac{24}{18+j13} \text{ V 转换为极坐标形式为 } V_0 = 1.08 \angle -35.84^\circ \text{ V}$$

13.11

$$I_s = 4\angle 0^\circ \text{ A}, V_s = 110\angle 30^\circ \text{ V}$$

$$R_1 = 200\Omega, Z_{L1} = j\omega L_1 = j(600 \times 0.8) = j480\Omega, Z_{L2} = j\omega L_2 = j(600 \times 1.2) = j720\Omega,$$

$$Z_M = j\omega M = j(600 \times 0.6) = j360\Omega, Z_C = \frac{1}{j\omega C} = \frac{1}{j(600 \times 12 \times 10^{-6})} = -j\frac{10^3}{7200} = -j138.89\Omega$$

$$R_2 = 150\Omega$$

设左侧网孔为 I_1 ，中间为 I_2 ，右侧为 I_3 ，即 $I_x = I_2 - I_3$

则对中间网孔，有：

$$200(I_2 - 4) + [j480I_1 + j360(I_2 - I_1)] + [j720(I_2 - I_3) - j360I_1] = 0$$

$$\text{化简得 } (200 + j480)I_2 - j360I_3 = 800 \quad (1)$$

右侧网孔，

$$[j720(I_3 - I_2) + j360I_2] + (150 - j138.89)I_3 + 110\angle 30^\circ = 0$$

$$\text{化简得 } -j360I_2 + (150 + j581.11)I_3 = -110\angle 30^\circ \quad (2)$$

联立 (1)(2)，可解得

$$\begin{cases} I_2 = 2.031 \angle -50.36^\circ \text{ A} = 1.296 - j1.564 \text{ A} \\ I_3 = 0.859 - j0.584 \text{ A} \end{cases}$$

$$\text{则 } I_x = I_2 - I_3 = (1.296 - j1.564) - (0.859 - j0.584) = 0.437 - j0.980 \text{ A}$$

转换为时域：

$$i_x = 1.07 \cos(600t - 66^\circ) \text{ A}$$

13.33

初级回路自阻抗 Z_{11} : $Z_{11} = 10 + j12 \Omega$ 次级回路自阻抗 Z_{22} : $Z_{22} = 20 + j40 + (-j5) = 20 + j35 \Omega$ 互感阻抗 Z_m : $Z_m = j\omega M = j15 \Omega$ 引入阻抗, $Z_r = \frac{(\omega M)^2}{Z_{22}}$, 代入得 $Z_r = \frac{15^2}{20 + j35} = \frac{225}{20 + j35}$ 则 $Z_{in} = Z_{11} + Z_r = 10 + j12 + \frac{225}{20 + j35}$

$$= 10 + j12 + \left(\frac{36}{13} - j\frac{63}{13} \right)$$

$$= 12.769 + j7.154 \Omega$$

13.53

左侧视为戴维南等效电路:

$$V_s = I_s \cdot R_p = 15 \times 5 = 75 \angle 0^\circ \text{ V}$$

$$R_{Th} = 5 + 3 = 8 \Omega$$

(a)

$$R_L' = \frac{R_L}{n^2} = \frac{200}{n^2}, \quad R_L' = R_{Th}, \quad \frac{200}{n^2} = 8$$

解得 $n=5$, 匝数比 $n=5$ 时传递的负载功率最大

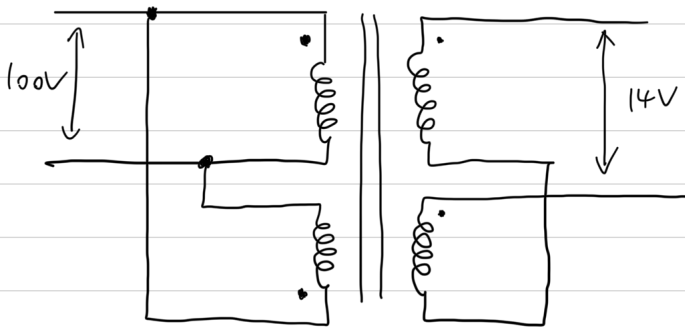
(b)

$$R_L' = \frac{200}{10^2} = 2 \Omega, \quad I_1 = \frac{V_{Th}}{R_{Th} + R_L'} = \frac{75}{8 + 2} = 7.5 \text{ A (rms)}$$

$$P_{load} = I_1^2 R_L' = 0.5^2 \cdot 2 = 112.5 \text{ W}$$

13.93

(a)



(b)

